

Navier–Stokes source result, exact propagation and the complete finite coupled bridge

This cumulative reader preserves the complete operator, moving-frame, physical-curl and signed-covariance calculations of the prior bridge. It adds the precise released theorem as source input, proves its exact orientation and retained-viscosity transports, and computes the two radial residual moments of the actual finite coupled candidate.

The source-selected positive swirl and its proved reflected negative counterpart are both retained with their corresponding forces. The source’s complete infinite construction is attributed to that manuscript; the distinct coupled candidate retains its explicit remainder. The companion 102-page reader contains the finite original-stage proofs.

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Part I

Complete finite operator, pulse and covariance bridge

1 Exact operator bridge to the released Navier–Stokes profiles

This bounded comparison uses the locally saved released Navier–Stokes source (the source pages are 7–15 and 24–27 for the profile, stress and pulse construction). Its stated physical viscosity is denoted by $\nu_{\text{NS}} > 0$. The Boussinesq lane variables and physical viscosity remain d, μ and $\nu = \sqrt{A_0}$; none is identified with ν_{NS} .

Use right-handed cylindrical coordinates (r, ϑ, z) and define

$$y_1 = z, \quad y_2 = \frac{r^2}{2} > 0, \quad v = (u^z, ru^r) = J \nabla_y \Psi, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The meridional divergence identity is $\partial_{y_1} v_1 + \partial_{y_2} v_2 = 0$. Define the circulation and meridional vorticity variables

$$\Gamma = ru^\vartheta, \quad \xi = \frac{\omega^\vartheta}{r}, \quad h = \frac{\Gamma}{2y_2} = \frac{u^\vartheta}{r}, \quad Q = \partial_{y_1}^2 + 2y_2 \partial_{y_2}^2, \quad M = Q + 4\partial_{y_2}.$$

The azimuthal momentum and meridional vorticity rows of the cylindrical equations (with arbitrary smooth force components) become, on $y_2 > 0$,

$$D_v \Gamma = \nu_{\text{NS}} Q \Gamma + r f^\vartheta, \tag{1}$$

$$D_v \xi = \nu_{\text{NS}} M \xi + \partial_{y_1} (h^2) + \text{curl}_y (f^z, f^r/r), \tag{2}$$

where $D_v = \partial_t + v \cdot \nabla_y$ and $\text{curl}_y(a, b) = \partial_{y_1} b - \partial_{y_2} a$.

Indeed, $\partial_r = r \partial_{y_2}$ and $\partial_r^2 + r^{-1} \partial_r = 2y_2 \partial_{y_2}^2 + 2\partial_{y_2}$. For Γ , the azimuthal viscous operator is $\partial_r^2 - r^{-1} \partial_r + \partial_z^2 = Q$. For $r\xi = \omega^\vartheta$, the cylindrical operator $(\partial_r^2 + r^{-1} \partial_r - r^{-2} + \partial_z^2)(r\xi)$, divided by r , is $M\xi$; explicitly the radial terms are $2y_2 \xi_{y_2 y_2} + 4\xi_{y_2}$. The centrifugal term is $r^{-2} \partial_z((u^\vartheta)^2) = \partial_{y_1}(\Gamma^2/(4y_2^2)) = \partial_{y_1}(h^2)$. These calculations prove every coefficient in (1)–(2), including the r force conversion.

The velocity–vorticity relation is also retained. Since $u^z = -\Psi_{y_2}$ and $u^r = \Psi_{y_1}/r$,

$$\xi = \frac{\partial_z u^r - \partial_r u^z}{r} = \frac{\Psi_{y_1 y_1}}{2y_2} + \Psi_{y_2 y_2} = \frac{Q\Psi}{2y_2}.$$

Thus the two transport rows do not by themselves replace the elliptic velocity reconstruction. Conversely, on a simply connected meridional domain, a smooth Ψ, Γ satisfying this relation and both transport rows recovers u . Define its momentum residual without pressure as $a = \partial_t u + (u \cdot \nabla)u - \nu_{\text{NS}} \Delta u - f$. The circulation row is $a^\vartheta = 0$, while the vorticity row is $\partial_z a^r - \partial_r a^z = 0$. The one-form $a^r dr + a^z dz$ is therefore closed, and $p = -\int (a^r dr + a^z dz)$ along a path from a fixed base point is path independent. Its derivatives are $p_r = -a^r$, $p_z = -a^z$. This proves the local momentum reconstruction, with pressure unique up to a function of time; boundary data are not removed.

The change from circulation to swirl rate is invertible on the exact retained domain: $\Gamma = 2y_2 h$. Product differentiation gives

$$Q(2y_2 h) = 2y_2 M h, \quad D_v(2y_2 h) = 2y_2 D_v h + 2v_2 h.$$

Therefore (1) is equivalent to

$$D_v h = \nu_{\text{NS}} M h - \frac{v_2}{y_2} h + \frac{f^\vartheta}{r}. \quad (3)$$

The source profile uses precisely this relation: in its similarity variables, E is the azimuthal velocity profile and $H_{\text{src}} = \sqrt{2X}E$ is the scaled circulation profile (source equations (4.3), (4.8)). Since $r = \sqrt{2qX}$ and $u^\vartheta = q^{-A}E$, its physical circulation is $\Gamma = q^{1/2-A}H_{\text{src}} = q^{-h_{\text{src}}}H_{\text{src}}$, where the source exponents satisfy $A = 1/2 + h_{\text{src}}$. Here $q > 0$ is the geometric similarity coordinate and $h_{\text{src}} > 0$ is an exponent; neither is a signed amplitude scale. This is the exact scale dictionary; it does not identify H_{src} with the Boussinesq scalar amplitude.

The sign of the blowout amplitude must be kept separate from that positive geometric coordinate. The axisymmetric equations have the exact involution

$$(u^r, u^\vartheta, u^z, p; f^r, f^\vartheta, f^z) \longmapsto (u^r, -u^\vartheta, u^z, p; f^r, -f^\vartheta, f^z).$$

It sends (Γ, h) to $(-\Gamma, -h)$, leaves ξ and the meridional flow unchanged, and preserves every displayed equation: the only quadratic swirl term is $\partial_{y_1}(h^2)$, while the azimuthal equation is linear in (Γ, f^ϑ) . Consequently a source-selected positive profile $E > 0$ (source pp. 7–9) does not justify assuming that every blowout branch is positive; its reflected branch has $E < 0$, $\Gamma < 0$ and the same blowup magnitude. The exponent $q^{-h_{\text{src}}}$ controls the magnitude in either branch. The source's positive covariance coefficients and viscosity are separate construction choices, not a sign claim about the velocity amplitude.

For comparison with the lane's controlled scalar equation, set $B = h^2$. The globally smooth square equation, including its geometric and viscous terms, is

$$D_v B = \nu_{\text{NS}} M B - 2\nu_{\text{NS}} |\nabla h|_Q^2 - \frac{2v_2}{y_2} B + \frac{2h}{r} f^\vartheta, \quad |\nabla h|_Q^2 = h_{y_1}^2 + 2y_2 h_{y_2}^2. \quad (4)$$

This follows by multiplying (3) by $2h$ and using $M(h^2) = 2hMh + 2|\nabla h|_Q^2$. At zeros of h equation (4) remains smooth; dividing by B is unnecessary and would lose that domain. On a component where $B > 0$, the equivalent expression is $D_v B = \nu_{\text{NS}} M B - \nu_{\text{NS}} |\nabla B|_Q^2 / (2B) - 2v_2 B / y_2 + 2h f^\vartheta / r$.

The exact relation to the Boussinesq lane is consequently an operator correspondence, not a scalar conjugacy. The lane's retained-viscosity scalar row has $D_v \Theta = d\Delta \Theta + f_\Theta$ in Cartesian material coordinates. A direct identification $\Theta = B$ would require, on the same physical domain, both $M = \Delta$ after the nonlinear coordinate pullback and $-2v_2 B / y_2 - 2\nu_{\text{NS}} |\nabla h|_Q^2 + 2h f^\vartheta / r$ to be absorbed into one prescribed force. The first condition fails as an operator identity: $M = \partial_{y_1}^2 + 2y_2 \partial_{y_2}^2 + 4\partial_{y_2}$ has variable metric and drift, whereas the lane's Δ is the Euclidean Cartesian Laplacian. The geometric, gradient and force terms must therefore be retained or calculated in the particular source state. Nonconstant profiles alone would not prove that such terms never cancel. The exact bridge is (1)–(4) with the full residual terms retained. It neither claims that the lane's controls are the source pulses nor changes the source's pressure reconstruction.

For the meridional velocity, the source's pulse construction supplies the following further exact local dictionary (source pp. 11–15 and 74–77). If w is a divergence-free pulse and π its pressure increment, then

$$R(u_B + w, p_B + \pi) = R(u_B, p_B) + L_{u_B}(w, \pi) + \nabla \cdot (w \otimes w),$$

$$L_{u_B}(w, \pi) = \partial_t w + (u_B \cdot \nabla) w + (w \cdot \nabla) u_B - \nu_{\text{NS}} \Delta w + \nabla \pi, \quad \nabla \cdot w = 0.$$

For the moving local phase with wavevector n and transverse amplitude t_m , the source equation is

$$t'_m + \mathcal{K} t_m + m^2 d_N t_m + i k m n \pi_m = -f_m, \quad n \cdot t_m = 0, \quad d_N = \varepsilon k^2 |n|^2,$$

with the source's moving shear matrix $\mathcal{K} = \begin{pmatrix} 0 & -2F & 0 \\ 2F + RF_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}$. The transverse projection uses

$$A_\Phi = -\mathcal{K} + \frac{n(n^T \mathcal{K} - n'^T)}{|n|^2}.$$

These are exact source pulse equations, including viscosity and pressure projection. Their quadratic covariance is chosen to cancel the annular stress; the source explicitly retains curl, cutoff, and correction interactions. The Boussinesq lane's two-dimensional pair and its full profile residuals have an exact algebraic form, but no identification with this three-dimensional pulse equation follows without an additional embedding that supplies the source's cylindrical geometry, moving plane, covariance cone and correction cycle. That embedding is not asserted here.

The current bridge therefore proves a precise morphism of variables, operators, forces and pulse residuals and records the exact obstruction to a direct scalar identification. It establishes the next programme datum while preserving the unproved status of any singular limit or third return.

2 Scope and source equations

This note records one local, exact comparison. It does not identify the coupled Boussinesq stages with the released Navier–Stokes construction and makes no claim about a global Navier–Stokes solution. The source is the released PDF `openai_navier_stokes.pdf`, SHA-256 `8c8a94ad9ac824c8b605b9827cadf7beaca48bd10b380de3cfc872a2c37afa81`. The source statements used here are Section 7, equations (7.2)–(7.8), (7.12)–(7.22), and Section 9, equations (9.1)–(9.2), on PDF pages 74–75, 77–81, and 101–103. In the source notation, on one lifted pulse rectangle,

$$\Phi = p\theta + p_z Z/\varepsilon + x_0 R - v(pF + p_z G), \quad n = \nabla_* \Phi, \quad (5)$$

with

$$n = (x_0 - v(pF_R + p_z G_R), p/R, p_z - \varepsilon v(pF_Z + p_z G_Z)). \quad (6)$$

For a nonzero integer harmonic m , the exact principal pulse equations are

$$n \cdot t_m = 0, \quad t'_m + K t_m + m^2 d_N t_m + i k m n \pi_m = -f_m, \quad d_N = \varepsilon k^2 |n|^2, \quad (7)$$

where a prime is the pulse-coordinate derivative and

$$K = \begin{pmatrix} 0 & -2F & 0 \\ 2F + RF_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}, \quad A_\Phi = -K + \frac{n(n^T K - (n')^T)}{|n|^2}. \quad (8)$$

The source pressure coefficient is exactly

$$\pi_m = \frac{i}{k m |n|^2} (n \cdot K t_m - (n') \cdot t_m + n \cdot f_m). \quad (9)$$

For $m = 1$, a real homogeneous pulse t_1 has the source energy identity (7.22):

$$\frac{1}{2}(|t_1|^2)' = -g \cdot (t_{1,r}(t_{1,\theta}, t_{1,z})) - \varepsilon k^2 |n|^2 |t_1|^2, \quad (10)$$

where $g = (RF_R, G_R)$ in the tangential coordinates. Thus the source pulse retains both shear transfer and viscous dissipation.

3 The coupled physical pair

For one activated layer of the coupled calculation, put

$$a_c = -\frac{J\zeta \cdot G_{<q}}{Kr^2}, \quad b_c = K\zeta_1, \quad \lambda_c^\Theta = \kappa K^2 r^2, \quad \lambda_c^\Omega = \varepsilon_c K^2 r^2. \quad (11)$$

Its exact retained-viscosity pair, including optional scalar and vorticity controls, is

$$\dot{y} = C_c y + c_c, \quad y = \begin{pmatrix} \Theta \\ \Omega \end{pmatrix}, \quad C_c = \begin{pmatrix} -\lambda_c^\Theta & a_c \\ b_c & -\lambda_c^\Omega \end{pmatrix}, \quad c_c = \begin{pmatrix} c_\theta \\ c_\omega \end{pmatrix}. \quad (12)$$

Here $G_{<q}$, ζ , and r evolve with the inherited background and phase ODEs; in particular no fixed-background substitution is made. Let v be a source pulse coordinate on an interval and let $q_v = dt/dv$ describe its physical-time parametrization. The forward source parametrization has $q_v > 0$; the algebra below also retains a negative orientation without changing the amplitude sign. Define $y'(v) = q_v(C_c y + c_c)$. Here K, r are the coupled frequency and phase length, not the source carrier k and cylindrical radius R . The original A_0, λ_0, μ and time scale $\nu = \sqrt{A_0}$ are unchanged. The source chart parameter ε is not identified with either physical diffusivity.

4 Exact local residual map

Let $B(v) \in \mathbb{R}^{3 \times 2}$ be a C^1 tangent frame with full column rank and $n(v)^T B(v) = 0$, so its range is exactly $n(v)^\perp$. Let $B_\ell(v)$ be any left inverse, $B_\ell B = I_2$; then BB_ℓ acts as the identity on $n^\perp$. Let $L(v) \in \mathbb{R}^{2 \times 2}$ be any C^1 coefficient map and embed the coupled pair into a transverse source amplitude by

$$t(v) = B(v)L(v)y(v). \quad (13)$$

For an embedding with a non-tangent frame the transversality condition would be

$$n(v)^T B(v)L(v)y(v) = 0. \quad (14)$$

For the tangent frame used here, the condition holds for every L, y . Set

$$H(v) = B_\ell(v)(A_\Phi(v)B(v) - B'(v)). \quad (15)$$

Then the exact projected source residual of the embedded coupled pair is

$$\mathcal{E}_{L,c} := B_\ell \text{proj}_{n^\perp} [t' - A_\Phi t + m^2 d_N t + f_m] = [L' + q_v LC_c - HL + m^2 d_N L] y + q_v Lc_c \quad (16)$$

$$+ B_\ell \text{proj}_{n^\perp} f_m. \quad (17)$$

In particular, with $L = I$ and no added source, the defect is the explicit matrix expression

$$\mathcal{E}_{I,0} = (q_v C_c - H + m^2 d_N I_2)y. \quad (18)$$

This gives a local residual relation with every actual coefficient retained.

Proof. Differentiate (13): $t' = B'Ly + BL'y + BLy'$. Since $y' = q_v(C_c y + c_c)$, differentiate $n^T B = 0$ to obtain $n^T B' = -(n')^T B$. Also $n^T A_\Phi B = -(n')^T B$, directly from (8). Thus $A_\Phi B - B'$ has transverse range and equals BH . Applying B_ℓ to the transverse terms and $B_\ell \text{proj}_{n^\perp}$ to the force proves (17). More explicitly,

$$n^T(t' - A_\Phi t + m^2 d_N t) = 0, \quad n^T(t' - A_\Phi t + m^2 d_N t + f_m) = n^T f_m. \quad (19)$$

The unprojected force need not be transverse. The pressure below removes precisely this remaining normal component. $\square$

5 Pressure reconstruction and full three-dimensional residual

For any embedded transverse t and arbitrary f_m , define

$$\pi_{L,c} = \frac{i}{km|n|^2} (n \cdot Kt - (n') \cdot t + n \cdot f_m). \quad (20)$$

Then the full source residual satisfies the exact decomposition

$$\mathcal{R}_{L,c} := t' + Kt + m^2 d_N t + ikmn\pi_{L,c} + f_m \quad (21)$$

$$= \text{proj}_{n^\perp} (t' + Kt + m^2 d_N t + f_m) = B \mathcal{E}_{L,c}, \quad (22)$$

where the last equality uses the definition of H and the transverse condition. Thus the pressure removes precisely the normal component; it does not remove $\mathcal{E}_{L,c}$, the moving-frame, shear, diffusion, or coupled-background mismatch. If one elects to force the source pulse to obey the released equation, then the unique additional transverse forcing is

$$f_{m,\text{corr}} = -B \mathcal{E}_{L,c}, \quad (23)$$

added to the old f_m . The old $\mathcal{E}_{L,c}$ is evaluated before this addition, so the new projected residual is zero. Since $n^T f_{m,\text{corr}} = 0$, pressure is unchanged. This is an exact local forced relation, not an identification of the physical constructions.

6 Diffusion and energy comparison

The source diffusion multiplier is $m^2 d_N = m^2 \varepsilon k^2 |n|^2$ in (7). The coupled pair has the two retained physical losses λ_c^Θ and λ_c^Ω , which are generally unequal to $m^2 d_N$. Their signed discrepancy in (18) is explicitly

$$(m^2 d_N - q_v \lambda_c^\Theta) \Theta, \quad (m^2 d_N - q_v \lambda_c^\Omega) \Omega \quad (24)$$

when $H = 0$ and C_c is expanded entrywise; with general H the exact diagonal entries are those of $q_v C_c - H + m^2 d_N I_2$. No damping is dropped. If one compares nonnegative loss magnitudes instead, the corresponding cost gaps are $|m^2 d_N - q_v \lambda_c^\Theta|$ and $|m^2 d_N - q_v \lambda_c^\Omega|$. The required finite-interval coordinate map can be constructed explicitly. Let $I = [v_0, v_1]$, and let X_c, X_s be the identity-initial-value fundamental matrices for $D_c = q_v C_c$ and $D_s = H - m^2 d_N I_2$, respectively. For either continuous matrix D , define

$$X_D(v) = I_2 + \sum_{j \geq 1} \int_{v_0 \leq w_j \leq \dots \leq w_1 \leq v} D(w_1) \cdots D(w_j) dw_j \cdots dw_1.$$

With $M_D = \sup_I \|D\|$ and $T_I = v_1 - v_0$, the j th integral has norm at most $(M_D T_I)^j / j!$. Uniform convergence proves the integral equation $X_D = I_2 + \int_{v_0}^v D X_D$ and hence $X_D' = D X_D$. The same iteration for $Z' = -ZD$, $Z(v_0) = I_2$ gives $Z X_D = I_2$ by differentiation, proving invertibility on I . For a specified invertible L_0 , set

$$L = X_s L_0 X_c^{-1}, \quad L^{-1} = X_c L_0^{-1} X_s^{-1}.$$

This has norm at most $\|L_0\| e^{(M_c + M_s) T_I}$; its inverse has the same bound with L_0^{-1} . Differentiation proves

$$L' = H L - q_v L C_c - m^2 d_N L, \quad (25)$$

so the homogeneous embedded pair has zero projected residual. This proves an actual finite-interval conjugacy along the retained coefficient path. It does not make the source phase, pressure, cutoff, or nonlinear covariance equal to the coupled Boussinesq fields. For general controls c_c , the forced term is $q_v L c_c$ in (17).

For an arbitrary embedded field (possibly complex), the exact energy balance retains its full defect:

$$\begin{aligned} \frac{1}{2}(|t|^2)' = & -\operatorname{Re}(\bar{t}^T \mathbf{K} t) - m^2 d_N |t|^2 - \operatorname{Re}(\bar{t}^T f_m) \\ & + \operatorname{Re}(\bar{t}^T B \mathcal{E}_{L,c}). \end{aligned} \quad (26)$$

This follows by taking the real Hermitian product of (22) with t . Its pressure term vanishes because n is real and $n^T t = 0$. The $2F$ part of $\mathbf{K}$ is skew-symmetric, hence its real work is zero; the shear work equals $\operatorname{Re}(\bar{t}_r(g_\theta t_\theta + g_z t_z))$. Only for a zero residual, zero force, $m = 1$, and real t does this reduce to (10). The real coupled pair has

$$\frac{1}{2} \frac{d}{dt} |y|^2 = -\lambda_c^\Theta \Theta^2 - \lambda_c^\Omega \Omega^2 + (a_c + b_c) \Theta \Omega + c_\theta \Theta + c_\omega \Omega, \quad (27)$$

which is a two-component physical energy identity. Equations (17) and (27) exhibit exactly where the source's three-dimensional shear flux and the coupled pair's two-dimensional exchange differ.

7 Proven scope

The source uses a cylindrical, localized, divergence-free curl construction (Section 7.4 and Section 9, PDF pages 85–87 and 102–103); the displayed principal pulse equation is only one component of that construction. The comparison above proves an exact local map and its full pressure-projected residual for every finite interval on which the denominators and frame left inverse exist. It preserves source viscosity, moving phase normal, pressure, rotation/shear matrix, and all coupled physical diffusion coefficients. It does not prove that the coupled third-growth or steering stages produce the source's pulse phase Φ , covariance, cutoff tails, or smooth terminal force. Those remain separate bridge problems.

The sign conventions in this comparison are also explicit. The quantities $q_v > 0$, $\varepsilon > 0$, and $d_N = \varepsilon k^2 |n|^2 > 0$ record the chosen orientation of physical time and the positive viscous coefficient; they are not assumptions on the sign of a pulse or blowout amplitude. The equations (7)–(9) are linear in (t_m, f_m) , so the simultaneous replacement $(t_m, f_m, \pi_m) \mapsto (-t_m, -f_m, -\pi_m)$ is an exact reflected branch. Likewise, the local map (13) is sign-equivariant under $y \mapsto -y$ and $c_c \mapsto -c_c$, with the coefficients fixed at the actual path. This is a statement about the retained linear equations, not a sign symmetry of the entire coupled nonlinear system. The source covariance is quadratic, so reversing a wave leaves that covariance unchanged. Reversing the full Navier–Stokes velocity need not reverse its force, since $(u \cdot \nabla)u$ is even in u . The full-flow map that reverses axisymmetric swirl is a spatial reflection; its complete formula is proved in the accompanying covariance bridge. No sign of a blowout or endpoint is inferred from positive geometric scales.

8 Source objects and the finite calculation

This calculation uses the preserved 165-page supplied manuscript *Finite Time Blowup for Navier–Stokes*, source equations (7.23)–(7.42), (9.12)–(9.14). Its PDF SHA-256 is

The source pulse fields, auxiliary rectangles, phases, and leading covariance columns are the given objects; their global construction is source input. Every finite map asserted below is proved here. The existing coupled-stage proofs remain in the complete 102-page reader. The companion pulse note proves the exact frame residual and a constructed finite-interval intertwiner along its evolving coefficients. The source uses positive $q(z, t)$, band scales $Q_\beta = 2^{-\ell_\beta}$, $\varepsilon_\beta = Q_\beta^h$, $A = 1/2 + h$; none is set to one here or identified with the original coupled $A_0, \lambda_0, \mu, \nu = \sqrt{A_0}$ or its physical diffusivities.

All operations in this note are on a compact set inside $q > 0$ and the source active annulus. There are finitely many relevant bands and locally finite source labels. Supports of newly prescribed inputs are strictly inside the annulus, so no lower bound on a leading amplitude at a vanishing shell edge is assumed. Endpoint and shell-edge estimates are not inferred from compact-interior estimates.

9 Physical curl of the actual retained amplitude

Let $y = (\Theta, \Omega)^T$ be the actual smooth retained coupled solution, $y' = q_v(C_c y + c_c)$, with the coefficients of the companion pulse note. Use the smooth actual coefficients on the chosen slow/auxiliary patch; all harmonic coefficients and cutoffs are independent of θ in cylindrical components, and each carrier has nonzero integer angular frequency. For its source tangent frame B , smooth coefficient map L , and a fixed smooth real compact cutoff χ of this type, put

$$t_m = \chi B L y, \quad \mathcal{E}_\chi = \chi \{ [L' + q_v L C_c - H L + m^2 d_N L] y + q_v L c_c \} + \chi' L y, \quad H = B_\ell (A_\Phi B - B').$$

Here prime differentiates along the pulse coordinate; the other slow and auxiliary derivatives remain in the spatial formulas below. Smooth parameter dependence of the companion's constructed L follows directly from its integral equations on a fixed interval: for a nonzero parameter multi-index I ,

$$\partial^I X(v) = \int_{v_0}^v \left[D \partial^I X + \sum_{0 < J \leq I} \binom{I}{J} (\partial^J D) \partial^{I-J} X \right] dw.$$

The initial parameter derivatives vanish for identity initial data. Inductively the known forcing is smooth and bounded on the compact patch; the same convergent ordered-integral iteration solves this equation. This proves every finite mixed derivative of X, X^{-1}, L . Since $n^T B = 0$, $n^T t_m = 0$. Differentiation proves $t'_m - A_\Phi t_m + m^2 d_N t_m = B \mathcal{E}_\chi$. Thus even the pulse-cutoff term is explicit.

A physical version of the exact curl construction makes the intermediate map to momentum residual unambiguous. For each chosen harmonic let $\phi(x, t) = k_\beta m \Phi_\beta(x, t, Y(x, t))$ be its real evaluated phase, let $\xi = \nabla_x \phi \neq 0$, and let $a(x, t) = Q_\beta^{-A} t_m$ be the physical transverse amplitude. The source coordinate change is $r = \sqrt{Q_\beta} R$, $z = Q_\beta^D Z$, $1 - t = Q_\beta T$, with $D = 1/2 - h$ and normalized axial derivative $D_z = \varepsilon_\beta \partial_Z$. It gives $\xi = (k_\beta m / \sqrt{Q_\beta}) n_\Phi$; all auxiliary chain derivatives are included in n_Φ . Hence $\xi \cdot a = 0$. Define in Cartesian components

$$C = \frac{i \xi \times a}{|\xi|^2}, \quad \mathcal{A} = C e^{i\phi}, \quad \nabla \times \mathcal{A} = (a + r_a) e^{i\phi}, \quad r_a = \nabla \times C. \quad (28)$$

The vector triple-product identity gives $i \xi \times C = a$, proving the last equality. It also shows that the physical potential has exactly the source factor $Q_\beta^{1/2-A}$ relative to its chart coefficient $i n_\Phi \times t_m / (k_\beta m |n_\Phi|^2)$. Cartesian differentiation of C includes derivatives of the cylindrical basis;

the same terms are the R^{-1} terms of source (7.37)–(7.38). Take real parts and sum the finite retained harmonics. This constructs a smooth compact velocity w_y with $\nabla \cdot w_y = 0$.

Retain a finite smooth source state (u_*, p_*) , including its background, actual leading waves and their curl corrections, and the real physical pressure increments obtained from the companion pulse pressure formula with $f_m = 0$:

$$\pi_{\chi, m} = \frac{i}{k_\beta m |n|^2} (n^T \mathbf{K} t_m - (n')^T t_m), \quad \pi_y = \sum_{\beta, m} \operatorname{Re}(Q_\beta^{-2A} \pi_{\chi, m} e^{i\phi_{\beta, m}}).$$

These are smooth and compact wherever t_m is. This definition fixes the pressure map without assuming the pulse residual vanishes. Define the actual candidate residual

$$\begin{aligned} R_y &= R_{\nu_{\text{NS}}}(u_* + w_y, p_* + \pi_y), \\ R_{\nu_{\text{NS}}}(u, p) &:= \partial_t u + (u \cdot \nabla)u - \nu_{\text{NS}} \Delta u + \nabla p. \end{aligned} \quad (29)$$

This is a definition by explicit differentiated fields, not an assumed source term. Its exact expansion is

$$R_y = R_{\nu_{\text{NS}}}(u_*, p_*) + \partial_t w_y + (u_* \cdot \nabla)w_y + (w_y \cdot \nabla)u_* - \nu_{\text{NS}} \Delta w_y + \nabla \pi_y + (w_y \cdot \nabla)w_y.$$

Every cutoff, parent, phase, and curl derivative occurs in this formula. Physical ν_{NS} is retained and is distinct from every source time-scaling convention. Define $\Delta R_y = R_y - R_{\nu_{\text{NS}}}(u_*, p_*)$. We correct the new compact increment; this difference has compact support since every added term does. The baseline residual remains explicitly.

The averaging needed next is the source extended-field operation, not a spatial average of the evaluated field. Evaluate derivatives on the extended coefficients first by the complete chain rule. In a fixed chart set $\mathcal{D}_i = \partial_{x_i} + \sum_\alpha (\partial_{x_i} Y_\alpha) \partial_{Y_\alpha}$ and similarly $\mathcal{D}_t$. These are the pullbacks of commuting physical derivatives, so $[\mathcal{D}_i, \mathcal{D}_j] = 0$ on their lifted domain. For smooth periodic coefficients their auxiliary derivative terms have zero Haar average. In cylindrical components angular differentiation also differentiates the basis. This defines an extended residual $\tilde{R}_y$ whose evaluation is exactly (29). Average the tangential components of the increment at fixed slow variables:

$$F_2 = \langle \Delta \tilde{R}_{y, \theta} \rangle, \quad F_1 = \langle \Delta \tilde{R}_{y, z} \rangle, \quad \langle g \rangle = \int_{\mathbb{T}^2} \frac{1}{2\pi} \int_0^{2\pi} g \, d\theta \, dY. \quad (30)$$

These F_e are smooth and compact in the chosen physical annulus. Equations (28)–(30) are the explicit intermediate map from the retained coupled pair to a signed momentum residual. They do not identify it with the original two-dimensional Boussinesq force.

10 Radial moments, support, and the original chart scales

For each fixed (z, t) , choose $0 < r_- < r_+$ containing the compact supports of both F_e and the chosen bumps in the active annulus. The source bump is

$$b_e(r, z, t) = q^{-(e+1)/2} \hat{b}_e(r/\sqrt{q}), \quad \int_0^\infty x^e \hat{b}_e(x) \, dx = 1, \quad e = 1, 2,$$

with interior support. The change $r = \sqrt{q} x$ proves $\int r^e b_e \, dr = 1$ exactly. Define

$$M_e = \int_0^\infty r^e F_e(r) \, dr, \quad \sigma_e(r) = -r^{-e} \int_0^r s^e (F_e(s) - b_e(s) M_e) \, ds. \quad (31)$$

Then

$$(\partial_r + e/r)\sigma_e = -F_e + b_e M_e, \quad \int_0^\infty r^e (F_e - b_e M_e) dr = 0. \quad (32)$$

Differentiate $r^e \sigma_e$ to prove the first identity; integrate and use the bump normalization for the second. The integral is zero below r_- and above r_+ , so σ_e has compact smooth support in the same enclosing interval. A homogeneous solution is cr^{-e} , and a compact homogeneous solution has $c = 0$; thus the primitive is unique among compact solutions of its displayed equation. Moreover a compact solution of $(\partial_r + e/r)\sigma = -F_e$ exists if and only if $M_e = 0$: necessity follows by integrating the weighted derivative, and sufficiency follows from (31). The two moments are therefore the exact surviving finite-dimensional terms.

For $C_e = \int_{r_-}^{r_+} s^e ds$, $|M_e| \leq C_e \|F_e\|_\infty$ and

$$\|\sigma_e\|_\infty \leq r_-^{-e} C_e (\|F_e\|_\infty + \|b_e\|_\infty |M_e|).$$

If $G_e = F_e - b_e M_e$, successive radial derivatives obey the exact recurrence

$$\partial_r^{j+1} \sigma_e = -\partial_r^j G_e - e \sum_{i=0}^j \binom{j}{i} (-1)^i i! r^{-i-1} \partial_r^{j-i} \sigma_e.$$

This proves every finite radial derivative bound inductively. Parameter derivatives follow by differentiating the compact integral; all derivatives of q, b_e and the moving physical data are retained. On the present compact set q is bounded away from zero. No terminal uniform estimate is implied.

The source (9.12)–(9.14) uses physical $F_e = Q^{-2A-1/2} E_e^*$, where $E_2^* = \langle E_\theta \rangle_Y$ and $E_1^* = \langle E_z \rangle_Y$. For our fields define $E_e^* = Q^{2A+1/2} F_e$ in that same convention. With $r = \sqrt{Q}R$ and Q fixed in a chart, the exact conversions are

$$M_e = Q^{e/2-2A} M_e^*, \quad M_e^* = \int R^e E_e^* dR, \quad b_e^*(R) = Q^{(e+1)/2} b_e(\sqrt{Q}R), \\ \Sigma_e(R) = Q^{2A} \sigma_e(\sqrt{Q}R), \quad (\partial_R + e/R) \Sigma_e = -E_e^* + b_e^* M_e^*. \quad (33)$$

Substitute $r = \sqrt{Q}R$ including its measure in the moment, and differentiate Σ_e to prove each factor. The input F_e is already auxiliary independent. Taking only an averaged moment of an unaveraged $F(R, Y)$ would not suffice: $F = b_e(R) \cos(2\pi Y_1)$ has zero averaged moment but a nonzero exterior primitive at each generic Y .

11 The source covariance derivative and its signed right inverse

At each slow point let $w_{\text{tan}} = (w_\theta, w_z)$ and define

$$\mathcal{C}(w) = \langle w_r w_{\text{tan}} \rangle, \quad \mathcal{B}(w, v) = \langle w_r v_{\text{tan}} + v_r w_{\text{tan}} \rangle.$$

Direct expansion proves $\mathcal{C}(w + v) = \mathcal{C}(w) + \mathcal{B}(w, v) + \mathcal{C}(v)$ and $D\mathcal{C}(w)[v] = \mathcal{B}(w, v)$.

Use the exact source real pulses $b_\sigma = \chi_g(\xi_g) \psi(v) t_\sigma^h \cos(k\Phi_\sigma)$, $\sigma \in \{+, -\}$. They exclude the slow cutoff η_β and scalar amplitudes; χ_g is an auxiliary transverse cutoff. The two signs have disjoint auxiliary supports. Put $H_{\text{cov}} = [\mathcal{C}(b_+) \mid \mathcal{C}(b_-)]$. For slow scalar coefficients independent of both averages,

$$\mathcal{C}(a_+ b_+ + a_- b_-) = H_{\text{cov}} (a_+^2, a_-^2)^T.$$

All cross products vanish by disjointness. The factor $1/2$ from $\langle \cos^2(k\Phi_\sigma) \rangle_\theta$ is already inside the columns.

On the source interior cone the fixed primary amplitudes satisfy $a_\sigma = \sqrt{(H_{\text{cov}}^{-1}T_{0,*})_\sigma} > 0$, $T_{0,*} = (Q/q)^{A+1/2}T_0$, and $W_0 = \sqrt{\varepsilon} \sum_\sigma a_\sigma b_\sigma$. For any signed slow real vector Σ , define

$$d_\Sigma = H_{\text{cov}}^{-1}\Sigma/\varepsilon, \quad \delta a_\sigma = (d_\Sigma)_\sigma/(2a_\sigma), \quad L_\Sigma = \sqrt{\varepsilon} \sum_\sigma \delta a_\sigma b_\sigma. \quad (34)$$

It follows exactly that

$$\mathcal{B}(W_0, L_\Sigma) = \varepsilon H_{\text{cov}}(2a \odot \delta a) = \Sigma, \quad (35)$$

$$\mathcal{C}(W_0 + L_\Sigma) = \varepsilon T_{0,*} + \Sigma + \varepsilon H_{\text{cov}}(\delta a_+^2, \delta a_-^2)^T. \quad (36)$$

This proves a linear right inverse of the derivative. It is not an exact inverse of the quadratic covariance map. There is no sign or size restriction on Σ in these identities; no square root of $H_{\text{cov}}^{-1}(T_{0,*} + \Sigma/\varepsilon)$ is taken.

With Euclidean and induced operator norms and $a_{\min} = \min_\sigma |a_\sigma| > 0$, they give the explicit bound

$$\|\mathcal{C}(L_\Sigma)\| \leq \frac{\|H_{\text{cov}}\| \|H_{\text{cov}}^{-1}\|^2}{4\varepsilon a_{\min}^2} \|\Sigma\|^2. \quad (37)$$

Indeed $\|\delta a\| \leq \|H_{\text{cov}}^{-1}\| \|\Sigma\|/(2\varepsilon a_{\min})$ and $\|(\delta a_\sigma^2)_\sigma\|_2 \leq \|\delta a\|_2^2$. All higher derivative costs are explicit from Leibniz and

$$\partial^I H_{\text{cov}}^{-1} = -H_{\text{cov}}^{-1} \sum_{0 < J \leq I} \binom{I}{J} (\partial^J H_{\text{cov}}) \partial^{I-J} H_{\text{cov}}^{-1},$$

and, for any nonzero scalar a , $\partial^I(a^{-1}) = -a^{-1} \sum_{0 < J \leq I} \binom{I}{J} (\partial^J a) \partial^{I-J}(a^{-1})$. Together with (34) these recurrences give finite bounds for each order without assuming constant amplitudes or constant columns.

12 Assembly, the actual covariance change, and residual cancellation

For the physical target $\sigma = (\sigma_2, \sigma_1)$ from (31), use the source squared slow partition $\sum_\beta \eta_\beta^2 = 1$ on its support. Set

$$\begin{aligned} W_{0,\text{phys}} &= \sum_\beta Q_\beta^{-A} \eta_\beta W_{0,\beta}, \\ v_{\text{pr}} &= \sum_\beta Q_\beta^{-A} \eta_\beta L_\beta(Q_\beta^{2A} \sigma). \end{aligned} \quad (38)$$

Overlapping slow labels have disjoint auxiliary supports. Taking cross covariance and using (35) gives

$$\mathcal{B}(W_{0,\text{phys}}, v_{\text{pr}}) = \sum_\beta Q_\beta^{-2A} \eta_\beta^2 Q_\beta^{2A} \sigma = \sigma.$$

Every band scale remains distinct. Likewise the leading physical factor is $Q^{-2A} \varepsilon (Q/q)^{A+1/2} = Q^{-A+h+1/2} q^{-A-1/2} = q^{-A-1/2}$ because $A = 1/2 + h$.

Apply the exact curl (28) to each coefficient of v_{pr} , including the slow cutoffs before curling. Write the resulting divergence-free increment as $v_s = v_{\text{pr}} + r_s$. Let $t_{s,m}$ denote the chart transverse harmonic coefficients of $\eta_\beta L_\beta(Q_\beta^{2A} \sigma)$, before multiplication by Q_β^{-A} . For each such coefficient

choose $\pi_{s,m} = i(n^T K t_{s,m} - (n')^T t_{s,m}) / (k_\beta m |n|^2)$ and sum $\text{Re}(Q_\beta^{-2A} \pi_{s,m} e^{i\phi_{\beta,m}})$ to define the smooth compact π_s . Its full gradients remain below. Let the actual old wave w consist of the wave part of u_* plus the candidate w_y , including all existing curl corrections, and write $w = W_{0,\text{phys}} + e_w$. Define tensor covariance $\mathcal{T}(w) = \langle w \otimes w \rangle$ and $\mathcal{B}_T(u, v) = \langle u \otimes v + v \otimes u \rangle$. Expansion, with $e_w = w - W_{0,\text{phys}}$, proves

$$\Delta \mathcal{T} = \mathcal{B}_T(W_{0,\text{phys}}, v_{\text{pr}}) + \mathcal{B}_T(e_w, v_{\text{pr}}) + \mathcal{B}_T(w, r_s) + \mathcal{T}(v_s). \quad (39)$$

Its first radial–tangential pair is σ . All other entries and the remaining three tensors are retained. For example their radial–tangential pair has bound

$$2\|e_w\|_{L_{\theta,Y}^2} \|v_{\text{pr}}\|_{L_{\theta,Y}^2} + 2\|w\|_{L_{\theta,Y}^2} \|r_s\|_{L_{\theta,Y}^2} + \|v_s\|_{L_{\theta,Y}^2}^2,$$

by Cauchy–Schwarz. The compact-set estimates above and the explicit curl differentiate to bound all these terms at each finite order.

For clarity the complete residual after adding v_s and its selected pressure π_s is, without any averaging,

$$\begin{aligned} R_{\nu_{\text{NS}}}(u + v_s, p + \pi_s) &= R_{\nu_{\text{NS}}}(u, p) + \partial_t v_s + (u \cdot \nabla) v_s + (v_s \cdot \nabla) u \\ &\quad - \nu_{\text{NS}} \Delta v_s + \nabla \pi_s + (v_s \cdot \nabla) v_s. \end{aligned} \quad (40)$$

This follows by expanding the product and retains viscosity and every pressure term. For a symmetric tensor S the averaged tangential divergence in physical cylindrical coordinates is

$$(\langle \nabla \cdot S \rangle)_\theta = (\partial_r + 2/r) \langle S_{r\theta} \rangle + \partial_z \langle S_{z\theta} \rangle, \quad (\langle \nabla \cdot S \rangle)_z = (\partial_r + 1/r) \langle S_{rz} \rangle + \partial_z \langle S_{zz} \rangle.$$

To verify, write the Cartesian divergence in the orthonormal cylindrical basis, using $\partial_\theta e_r = e_\theta$, $\partial_\theta e_\theta = -e_r$; periodic angular derivatives average to zero and the two basis terms give $2/r$ and $1/r$ respectively. The auxiliary chain terms average to zero by periodicity, but the physical slow derivatives remain.

Let $U = u_* + w_y$, $P_y = p_* + \pi_y$, and set

$$\mathcal{L} = \partial_t v_s + (U \cdot \nabla) v_s + (v_s \cdot \nabla) U - \nu_{\text{NS}} \Delta v_s + \nabla \pi_s.$$

For a fully explicit averaged remainder, write $U = u_{\text{mean}} + w$, where u_{mean} is its angular mean, and put $\mathcal{V} = \mathcal{L} - (w \cdot \nabla) v_s - (v_s \cdot \nabla) w$. Then the average of (40) is $\langle R_y \rangle + \langle \mathcal{V} \rangle + \nabla \cdot \Delta \mathcal{T}$. Let $S_0 = \mathcal{B}_T(W_{0,\text{phys}}, v_{\text{pr}})$ and $S_{\text{rem}} = \mathcal{B}_T(e_w, v_{\text{pr}}) + \mathcal{B}_T(w, r_s) + \mathcal{T}(v_s)$. Consequently the radial divergence of the first pair in (39) changes the prescribed averaged residual F_e into $b_e M_e$ by (32). The axial tensor divergences, the other three tensors, the exact linear terms and the pressure terms of (40) remain in the new residual. Equivalently the new averaged tangential residual is exactly

$$\begin{aligned} \langle R_{\text{new}} \rangle_{\text{tan}} &= \langle R_{\nu_{\text{NS}}}(u_*, p_*) \rangle_{\text{tan}} + (b_2 M_2, b_1 M_1) + \langle \mathcal{V} \rangle_{\text{tan}} \\ &\quad + \partial_z ((S_0)_{z\theta}, (S_0)_{zz}) + (\nabla \cdot S_{\text{rem}})_{\text{tan}}. \end{aligned}$$

Every term here is defined by the displayed differentiated fields and tensors. None is assumed to vanish. The map from y through (30), (31), (34), and (38) is now an actual composite construction. It retains the signed residual produced by the coupled candidate and the source’s original physical and chart units.

13 Signs and the exact full-flow reflection

The maps $F_e \mapsto M_e \mapsto \sigma_e \mapsto L_\Sigma$ are linear and reverse sign with their inputs; $\mathcal{C}(L_\Sigma)$ is even. The two primary amplitudes may also be replaced independently by $s_\sigma a_\sigma$, $s_\sigma \in \{-1, 1\}$, provided the same fixed choice is used in each denominator: the cross identity is unchanged. This does not negate the nonlinear Navier–Stokes equation.

Here is the exact full-flow map. In Cartesian coordinates let $O = \text{diag}(1, -1, 1)$ and define

$$u^\sharp(x, t) = Ou(Ox, t), \quad p^\sharp(x, t) = p(Ox, t), \quad f^\sharp(x, t) = Of(Ox, t).$$

The chain rule gives $\nabla u^\sharp = O(\nabla u)(Ox, t)O$, $\Delta u^\sharp = O(\Delta u)(Ox, t)$, and $(u^\sharp \cdot \nabla)u^\sharp = O((u \cdot \nabla)u)(Ox, t)$. Thus $\nabla \cdot u^\sharp = 0$ and $R_{\nu_{\text{NS}}}(u^\sharp, p^\sharp) = OR_{\nu_{\text{NS}}}(u, p)(Ox, t)$ with the same positive viscosity. In cylindrical components this is

$$(u_r^\sharp, u_\theta^\sharp, u_z^\sharp)(r, \theta, z, t) = (u_r, -u_\theta, u_z)(r, -\theta, z, t),$$

and the force transforms identically. Pressure reflects in its argument. For an axisymmetric field this reverses exactly its swirl; meridional components remain fixed. The entire momentum tensor is mapped to OTO^T , so its radial–tangential pair transforms by $(T_{r\theta}, T_{rz}) \mapsto (-T_{r\theta}, T_{rz})$ after angular averaging. Vorticity transforms as $\omega^\sharp(x, t) = \det(O)O\omega(Ox, t)$, obtained by applying the same derivative identities to the Levi-Civita formula for curl. The map is an involution, preserves the kinetic-energy integral (by $|\det O| = 1$), compact support, smoothness, and blowup magnitude along the reflected path. It preserves zero initial data when present. It proves transport of any established axisymmetric signed swirl divergence to the opposite sign; it does not itself prove an endpoint.

The source five-moment correction, auxiliary mean inverse, and infinite summation are subsequent source operations. Their full estimates are not re-proved here. The earlier draft formulas for them are withdrawn: in particular the fifth moment row is $\int (2RG\gamma_d - RV\Delta v) \, dR = -J_z$, without an extra outer R , and its reserved profile in a Q chart includes $(Q/q)^A$. The finite map proved above retains the two moments and all other residual terms, so it establishes neither a third shooting return nor an infinite modified viscous sequence.

Part II

The released result and exact propagation

14 The released result and its actual signed path

The source in this section is the supplied 166-page edition of *Finite Time Blowup for Navier–Stokes*, whose SHA-256 is

0e779481c4da40bd28d1e642e1d8ca57447d129610df28dfa5a11e9af8ae228f.

Its Theorem 1.1, on page 1, states the following result for the standard three-dimensional incompressible Navier–Stokes equation. For every fixed physical viscosity $\nu_{\text{NS}} > 0$, the source constructs

$$f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3), \quad u, p \in C^\infty(\mathbb{R}^3 \times [0, 1)),$$

with a fixed compact spatial support K for u and p , such that

$$\begin{aligned} \partial_t u + (u \cdot \nabla)u - \nu_{\text{NS}} \Delta u + \nabla p &= f, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = 0, \\ \sup_{t < 1} \|u(t)\|_2 &< \infty, \quad \limsup_{t \uparrow 1} \|u(t)\|_\infty = \infty. \end{aligned} \tag{41}$$

It also excludes a global smooth solution with the same force and datum and uniformly bounded kinetic energy. Its Corollary 10.6, pages 125–126, gives a periodic construction on $\mathbb{R}^3/\mathbb{Z}^3$ with smooth time-compact force, zero initial datum and no global smooth continuation. The source identifies these with alternatives (C) and (D) in the referenced Millennium problem statement. The force is nonzero, as the source explicitly notes on page 124. These are statements of the cited source theorem; the finite algebra checks in this lane do not constitute an independent verification of the complete 166-page proof.

The source proves more than an unsigned norm divergence. In its viscosity-one construction, equations (10.20)–(10.21), page 124, give

$$\begin{aligned} A &= \tfrac{1}{2} + h, \quad 0 < h < 1/100, \quad X_{\text{in}} > 0, \quad e_0 > 0, \\ x_\tau &= (\sqrt{2X_{\text{in}}\tau}, 0, 0), \quad t_\tau = 1 - \tau, \\ u_\theta(x_\tau, t_\tau) &= \tau^{-A}(e_0 + \epsilon(\tau)), \quad |\epsilon(\tau)| \leq C\tau^{2h} \quad (0 < \tau < \tau_0). \end{aligned} \tag{42}$$

Here C, τ_0 are finite constants supplied by the source asymptotic; e_0 and X_{in} retain their source values. If $C > 0$, choose $0 < \tau_1 \leq \tau_0$ so that $C\tau_1^{2h} \leq e_0/2$; if $C = 0$, take $\tau_1 = \tau_0$. It follows directly that

$$u_\theta(x_\tau, t_\tau) \geq \frac{e_0}{2}\tau^{-A} \longrightarrow +\infty \quad (0 < \tau < \tau_1).$$

The observation point moves to the origin. This assertion does not replace that path by the fixed point $x = 0$. On the path, $q = \tau > 0$, and at its cylindrical angle $\theta = 0$ one has $e_\theta = e_2$. The geometric domain $q > 0$, the positive exponent h , and the signed velocity value are three retained parts of the same explicit formula.

15 Exact propagation at retained viscosities and orientations

For any smooth fields u, p define the physical momentum residual

$$R_{\nu_b}(u, p) = \partial_t u + (u \cdot \nabla)u - \nu_b \Delta u + \nabla p.$$

Let $\nu_b > 0$ and $\nu_t > 0$ be the base and target physical viscosities, and retain

$$\rho = \sqrt{\nu_t/\nu_b}, \quad O^T O = O O^T = I, \quad a \in \mathbb{R}^3.$$

Define the affine spatial coordinate $y = O^T(x - a)/\rho$ and the fields

$$u^\sharp(x, t) = \rho O u(y, t), \quad p^\sharp(x, t) = \rho^2 p(y, t), \quad f^\sharp(x, t) = \rho O f(y, t). \quad (43)$$

No time coordinate is changed. These formulas give a bijection on smooth field triples, with inverse

$$u(y, t) = \rho^{-1} O^T u^\sharp(a + \rho O y, t), \quad p(y, t) = \rho^{-2} p^\sharp(a + \rho O y, t), \quad f(y, t) = \rho^{-1} O^T f^\sharp(a + \rho O y, t).$$

For the Jacobian convention $(Du)_{ij} = \partial_j u_i$, differentiation gives, without dropping an orientation or viscosity,

$$\begin{aligned} Du^\sharp &= O(Du)(y, t)O^T, & \nabla \cdot u^\sharp &= (\nabla \cdot u)(y, t), \\ \partial_t u^\sharp &= \rho O \partial_t u(y, t), & (u^\sharp \cdot \nabla_x) u^\sharp &= \rho O((u \cdot \nabla_y) u)(y, t), \\ \Delta_x u^\sharp &= \rho^{-1} O \Delta_y u(y, t), & \nabla_x p^\sharp &= \rho O \nabla_y p(y, t). \end{aligned}$$

The nonlinear identity follows by multiplying $Du^\sharp$ by $u^\sharp$ and using $O^T O = I$. The Laplacian identity follows by contracting the two spatial derivative indices and using orthogonality. Since $\nu_t = \rho^2 \nu_b$, these identities prove

$$R_{\nu_t}(u^\sharp, p^\sharp) - f^\sharp = \rho O(R_{\nu_b}(u, p) - f)(y, t). \quad (44)$$

In particular the map preserves the exact equation as well as each nonzero residual with its exact factor.

For vorticity, the cross-product identity $(Ov) \times (Ow) = \det(O)O(v \times w)$ gives

$$\omega^\sharp(x, t) = \det(O)O\omega(y, t). \quad (45)$$

One proof of this identity takes its scalar product with Oz : both sides give $\det(O) \det[v, w, z]$ for every z . Applying it to the contracted first derivatives proves the vorticity formula; the factor ρ in velocity cancels the inverse factor in a derivative.

Spatial support becomes $a + \rho OK$. The Jacobian has absolute determinant ρ^3 , for both signs of $\det O$. Hence, for $1 \leq p < \infty$,

$$\|u^\sharp(t)\|_p = \rho^{1+3/p} \|u(t)\|_p, \quad \|u^\sharp(t)\|_\infty = \rho \|u(t)\|_\infty, \quad \|u^\sharp(t)\|_2^2 = \rho^5 \|u(t)\|_2^2. \quad (46)$$

The Frobenius norm of $O(Du)O^T$ equals that of Du . Consequently, with integrals over the same time interval,

$$\nu_t \int \|\nabla u^\sharp\|_2^2 dt = \rho^5 \nu_b \int \|\nabla u\|_2^2 dt.$$

For arbitrary directions $v_1, \dots, v_j$, the exact force derivative is

$$D_x^j \partial_t^m f^\sharp[v_1, \dots, v_j](x, t) = \rho^{1-j} O(D_y^j \partial_t^m f)[O^T v_1, \dots, O^T v_j](y, t).$$

Thus smoothness, compact space-time support and vanishing of every derivative at a terminal point are preserved. The initial datum remains zero. A global smooth bounded-energy competitor for the transformed datum and force would, under the displayed inverse, be a competitor for the base datum and force, because its energy is multiplied by the fixed factor ρ^{-5} . This proves propagation of the source noncontinuation conclusion through the map.

16 The source positive branch and the exact negative branch

Apply the map to the source field in (42). Here $\nu_b = 1$ specifies the source object being used; physical target viscosity ν_t remains arbitrary and $\rho = \sqrt{\nu_t}$ is retained. Take $a = 0$ and the two particular orthogonal matrices

$$O_+ = I, \quad O_- = \text{diag}(1, -1, 1).$$

Both fix e_1 , so they give the same spatial observation path $x_\tau^\sharp = \rho x_\tau$ and the same time $1 - \tau$. Since $O_+ e_2 = e_2$ and $O_- e_2 = -e_2$, their exact signed values are

$$u_{\pm, \theta}^\sharp(\rho x_\tau, 1 - \tau) = \pm \rho \tau^{-A}(e_0 + \epsilon(\tau)) \longrightarrow \begin{cases} +\infty, & +, \\ -\infty, & -. \end{cases} \quad (47)$$

All support, energy, force and viscosity statements in the preceding section apply to both triples. For the negative branch the force is $\rho O_- f(O_- x / \rho, t)$. Equality of this force with the positive branch force is not needed and is not asserted.

For a general, possibly nonaxisymmetric field the reflection alone has the exact cylindrical component formula

$$(u^r, u^\theta, u^z)^\sharp(r, \theta, z, t) = (u^r, -u^\theta, u^z)(r, -\theta, z, t).$$

Indeed $O_- e_r(-\theta) = e_r(\theta)$, $O_- e_\theta(-\theta) = -e_\theta(\theta)$ and $O_- e_z = e_z$. This proves the formula even in the presence of all oscillatory corrections. For the axisymmetric background the argument $-\theta$ can be dropped because its components are independent of θ . For a velocity covariance tensor the exact Cartesian transformation is $T^\sharp(x, t) = \rho^2 O T(y, t) O^T$: expand the outer product of $\rho O w$ with itself and use linearity of the retained average. For its angularly averaged symmetric tensor, the reflection basis calculation at $\rho = 1$ gives

$$(T_{r\theta}, T_{rz})^\sharp = (-T_{r\theta}, T_{rz}).$$

Under the additional viscosity dilation its components have factor ρ^2 at the corresponding points. Thus the two covariance columns and their target transform together. Positive coefficients in a source representation $T = c_1 v_1 + c_2 v_2$, $c_i > 0$, remain the same coefficients of the transformed columns; no change of coefficient sign is required to realize the negative-swirl branch.

The source's periodic construction also has this reflection: O_- preserves $\mathbb{Z}^3$, so $[x] \mapsto [O_- x]$ is well defined on $\mathbb{R}^3 / \mathbb{Z}^3$. The derivative identities apply to periodic lifts. This gives both signed orientations of the source Corollary 10.6. Arbitrary $O \in O(3)$ need not preserve this particular lattice; only the lattice-preserving reflection just specified is used for this periodic assertion.

17 Propagation into the coupled calculation

The preceding sections use the released whole-space and periodic constructions at the precise places where those source results apply. The original coupled Boussinesq work is due to Levent Alpöge and Tristan Buckmaster. This lane's retained coupled system, its finite controlled stages and its residual maps remain the separate objects defined in the preceding parts of this reader.

For every compact candidate $(u_* + w_y, p_* + \pi_y)$ in the covariance part, (44) propagates its full computed residual. Linearity of the transformation and the exact nonlinear identity give in particular

$$\Delta R_y^\sharp(x, t) = \rho O \Delta R_y(y, t).$$

Thus the finite candidate's cutoff forces, curl errors and covariance remainders all have an exact transformed counterpart. A nonzero remainder is carried to a nonzero remainder, because the map is invertible. The source's residual flatness at the singular point belongs to its completed local source construction; its extended smooth force need not vanish on the entire time-one slice. The next section calculates the surviving moments of the actual finite candidate, so that propagation continues through its computed terms rather than assuming that its different coupled coefficients inherit the source's infinite estimates.

The earlier suggestion of a negative blowout imposes no condition on the programme. Formula (42) is the source-selected sign; formula (47) proves the exact relation of its two orientations. Both formulas retain a positive physical viscosity and the source coordinate domain $q > 0$.

Part III

The actual coupled candidate radial moments

18 The two moments of the actual compact curl increment

The objects in this calculation are the finite smooth source state (u_*, p_*) and the actual added curl field $v = w_y$ with its selected pressure $\pi = \pi_y$ from the signed covariance bridge. Their physical residual increment is

$$\Delta R = \partial_t v + (u_* \cdot \nabla)v + (v \cdot \nabla)u_* + (v \cdot \nabla)v - \nu_{\text{NS}}\Delta v + \nabla\pi. \quad (48)$$

The physical viscosity ν_{NS} is unchanged. Both velocities are divergence free: the source state uses its actual divergence-free background and curl increments, and v is the curl of its specified compact potential. No residual equation for u_* is required in (48); its residual has been subtracted exactly. All radial supports involving v or π lie in a fixed compact subinterval of $(0, \infty)$ on the compact space-time patch at issue.

Here the source extended field is indispensable. Its absolute auxiliary coordinate is $Y \in \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, and physical evaluation uses precisely source (6.2)–(6.4):

$$\begin{aligned} J_g &= \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}, \quad \Lambda_g = 4 - \sqrt{2}, \quad T_g = 4 + \sqrt{2}, \quad b_g = \sqrt{2} - 1, \\ \mathbf{v}_r &= (1, -b_g), \quad \mathbf{v}_t = (b_g, 1), \quad \rho_g = \frac{\log \Lambda_g}{\log T_g}, \quad \kappa_s = 10^{-5}, \quad d_r = 2((1+h)\rho_g - h\kappa_s), \\ Y(r, t) &= \mathbf{v}_r r^{d_r} + \mathbf{v}_t t \pmod{\mathbb{Z}^2}, \quad \mathcal{D}_r = \partial_r + d_r r^{d_r-1} \mathbf{v}_r \cdot \partial_Y, \quad \mathcal{D}_t = \partial_t + \mathbf{v}_t \cdot \partial_Y. \end{aligned} \quad (49)$$

These are the source's original quantities, not the coupled physical frequency or diffusivity. Angular and axial physical derivatives are ∂_θ and ∂_z , because $Y(r, t)$ depends on neither variable. Every field below is first differentiated using these lifted physical derivatives. A tilde is suppressed for this extended representation. Evaluation at (49) recovers the physical field and its physical derivatives by the chain rule.

The incompressibility needed below holds at every independent Y . Indeed the actual finite source construction retains the form $u_* = \text{curl}_{\mathcal{D}} \mathcal{A}_* + B_* e_\theta$ with $\partial_\theta B_* = 0$, and the added field is $v = \text{curl}_{\mathcal{D}} \mathcal{A}_y$. The retained source potentials here include its finite mean and wave curl increments. The lifted derivatives in (49), together with ∂_θ and ∂_z , commute, and $\mathcal{D}_r r = 1$. Substitution in the cylindrical curl and divergence formulas therefore gives $\text{div}_{\mathcal{D}} \text{curl}_{\mathcal{D}} \mathcal{A} = 0$ identically in (r, θ, z, t, Y) : each mixed derivative cancels its opposite, including the derivative of r^{-1} . Also $\text{div}_{\mathcal{D}}(B_* e_\theta) = r^{-1} \partial_\theta B_* = 0$. Thus extended incompressibility is proved from the actual retained potential representation, rather than inferred from its restriction to the physical phase map.

Use the normalized averages

$$\langle a \rangle_\theta = \frac{1}{2\pi} \int_0^{2\pi} a \, d\theta, \quad \langle a \rangle_Y = \int_{\mathbb{T}^2} a \, dY, \quad \langle a \rangle = \langle \langle a \rangle_\theta \rangle_Y, \quad \int_{\mathbb{T}^2} 1 \, dY = 1.$$

They act on cylindrical components before physical evaluation, exactly as source (3.7)–(3.9). All claimed integral identities below concern these specified averages; the exact relation to evaluated moments is also proved below.

18.1 The actual harmonic curl has zero angular mean

For each retained source harmonic, its extended potential and pressure have cylindrical components of the form

$$C_j(r, z, t, Y)e^{i(n_j\theta + \psi_j(r, z, t, Y))}, \quad P_j(r, z, t, Y)e^{i(n_j\theta + \psi_j(r, z, t, Y))}, \quad n_j = k_{\beta_j} m_j p_j \in \mathbb{Z} \setminus \{0\}.$$

The coefficients include the actual cutoff χ , frame B , map L , coupled solution $y = (\Theta, \Omega)^T$, band powers and pressure formula. Their cylindrical components are independent of θ . In particular, no sign of Θ , Ω or the real part of a harmonic is prescribed. For a vector potential $\mathcal{A}$ the exact cylindrical curl is

$$(\text{curl } \mathcal{A})_r = r^{-1} \partial_\theta \mathcal{A}_z - \partial_z \mathcal{A}_\theta, \quad (\text{curl } \mathcal{A})_\theta = \partial_z \mathcal{A}_r - \mathcal{D}_r \mathcal{A}_z, \quad (\text{curl } \mathcal{A})_z = (\mathcal{D}_r + r^{-1}) \mathcal{A}_\theta - r^{-1} \partial_\theta \mathcal{A}_r.$$

These formulas include the derivatives of the cylindrical basis. Each displayed operation retains the same factor $e^{in_j\theta}$: the coefficients of $\mathcal{D}_r$ are independent of θ , and ∂_θ multiplies that factor by in_j . Integration of the factor over the angular circle gives zero. Taking real parts and the actual finite harmonic sum proves

$$\langle v_r \rangle_\theta = \langle v_\theta \rangle_\theta = \langle v_z \rangle_\theta = \langle \pi \rangle_\theta = 0 \quad \text{at every } (r, z, t, Y). \quad (50)$$

This includes every curl and cutoff correction, not merely the principal amplitude. The same identities hold after physical evaluation because $Y(r, t)$ is independent of θ .

For every smooth periodic coefficient a , integration of its auxiliary derivatives gives

$$\langle \mathcal{D}_r a \rangle = \partial_r \langle a \rangle, \quad \langle \mathcal{D}_t a \rangle = \partial_t \langle a \rangle, \quad \langle \partial_z a \rangle = \partial_z \langle a \rangle, \quad \langle \partial_\theta a \rangle = 0. \quad (51)$$

There is no omitted variable coefficient in this assertion: $d_r r^{d_r-1} \mathbf{v}_r$ is independent of Y . Applying the first identity again proves $\langle \mathcal{D}_r^2 a \rangle = \partial_r^2 \langle a \rangle$. Explicitly, if $c(r) = d_r r^{d_r-1} \mathbf{v}_r$, then

$$\mathcal{D}_r^2 a = \partial_r^2 a + 2c \cdot \partial_Y \partial_r a + c' \cdot \partial_Y a + (c \cdot \partial_Y)^2 a;$$

each of the three additional terms has zero auxiliary average.

18.2 Exact removal of the linear and mean-flow terms

Put

$$\bar{u}_* := \langle u_* \rangle_\theta, \quad w_* := u_* - \bar{u}_*.$$

The field $\bar{u}_*$ may retain auxiliary dependence; none is discarded. Since it is independent of θ , (50) gives

$$\langle \bar{u}_* \otimes v + v \otimes \bar{u}_* \rangle = 0.$$

Define the actual symmetric cross-and-quadratic tensor

$$S := \langle w_* \otimes v + v \otimes w_* + v \otimes v \rangle. \quad (52)$$

Products of all original source waves with the added curl field are retained in this formula, including old and new curl remainders.

For two divergence-free fields a, b , Cartesian component expansion gives

$$\nabla \cdot (a \otimes b + b \otimes a) = (b \cdot \nabla) a + (a \cdot \nabla) b, \quad \nabla \cdot (v \otimes v) = (v \cdot \nabla) v.$$

For example $\sum_j \partial_j (a_i b_j) = \sum_j b_j \partial_j a_i + a_i \sum_j \partial_j b_j$; the last term is zero. The lifted derivatives obey the same product rule, so the identities hold on the extended domain before evaluation. The three advection terms in (48) are therefore exactly the divergence of $u_* \otimes v + v \otimes u_* + v \otimes v$.

For completeness, set $\mathcal{L}_0 = \mathcal{D}_r^2 + r^{-1} \mathcal{D}_r + r^{-2} \partial_\theta^2 + \partial_z^2$. The vector Laplacian and pressure gradient in the two components are

$$(\Delta v)_\theta = (\mathcal{L}_0 - r^{-2})v_\theta + 2r^{-2} \partial_\theta v_r, \quad (\Delta v)_z = \mathcal{L}_0 v_z, \quad (\nabla \pi)_\theta = r^{-1} \partial_\theta \pi, \quad (\nabla \pi)_z = \partial_z \pi.$$

Equations (50)–(51) prove separately

$$\langle \mathcal{D}_t v \rangle_{\text{tan}} = 0, \quad -\nu_{\text{NS}} \langle \Delta v \rangle_{\text{tan}} = 0, \quad \langle \nabla \pi \rangle_{\text{tan}} = 0. \quad (53)$$

Thus viscosity and pressure were included before averaging; their vanishing here is a proved consequence of the actual carriers.

If T is symmetric, differentiation of $e_r = (\cos \theta, \sin \theta, 0)$, $e_\theta = (-\sin \theta, \cos \theta, 0)$ gives

$$\begin{aligned} (\nabla \cdot T)_\theta &= (\mathcal{D}_r + 2/r)T_{r\theta} + r^{-1} \partial_\theta T_{\theta\theta} + \partial_z T_{z\theta}, \\ (\nabla \cdot T)_z &= (\mathcal{D}_r + 1/r)T_{rz} + r^{-1} \partial_\theta T_{\theta z} + \partial_z T_{zz}. \end{aligned}$$

Indeed $\partial_\theta e_r = e_\theta$ and $\partial_\theta e_\theta = -e_r$ supply the two radial connection terms in the first line and the single radial connection term in the second. Averaging these formulas and using (51), (52), and (53) proves the exact residual identities

$$F_2 := \langle \Delta R_\theta \rangle = (\partial_r + 2/r)S_{r\theta} + \partial_z S_{z\theta}, \quad F_1 := \langle \Delta R_z \rangle = (\partial_r + 1/r)S_{rz} + \partial_z S_{zz}. \quad (54)$$

In particular every mean-flow cross term vanishes as the divergence of its zero averaged tensor, even when the mean depends on Y .

18.3 The surviving axial fluxes and their signs

Define the two moments with the original physical radial measure:

$$M_2(z, t) = \int_0^\infty r^2 F_2(r, z, t) dr, \quad M_1(z, t) = \int_0^\infty r F_1(r, z, t) dr. \quad (55)$$

Their exact values are

$$M_2 = \partial_z J_2, \quad J_2 = \int_0^\infty r^2 \langle w_{*,z} v_\theta + v_z w_{*,\theta} + v_z v_\theta \rangle dr, \quad (56)$$

$$M_1 = \partial_z J_1, \quad J_1 = \int_0^\infty r \langle 2w_{*,z} v_z + v_z^2 \rangle dr. \quad (57)$$

To prove them, multiply (54) by the respective weight: $r^2(\partial_r + 2/r)S_{r\theta} = \partial_r(r^2 S_{r\theta})$ and $r(\partial_r + 1/r)S_{rz} = \partial_r(r S_{rz})$. The weighted boundary values at both ends of $(0, \infty)$ are zero, because each entry of S contains at least one compactly supported factor v . Differentiation under the radial integral is valid by the same compact support and smoothness. This proves (56)–(57) including their signs and factors.

There is no pointwise cancellation of the axial derivatives in these formulas. Since the added field has compact axial support, J_1 and J_2 have compact axial support and

$$\int_{\mathbb{R}} M_e(z, t) dz = 0 \quad (e = 1, 2). \quad (58)$$

This follows by evaluating the boundary values of J_e at the two axial ends. It does not assert $M_e(z, t) = 0$ at an individual axial point. The compact radial primitive in the covariance bridge must therefore retain its $b_e M_e$ terms with these actual M_e .

At fixed source state, replacing the added field and pressure by their negatives sends the old-wave cross terms in (56)–(57) to their negatives and leaves the quadratic terms unchanged. Thus negating an amplitude is not a sign reversal of either full moment. Under the full physical reflection $O = \text{diag}(1, -1, 1)$, applied to the source state and added field together as $a^\sharp(x, t) = Oa(Ox, t)$, the cylindrical radial and axial components retain their signs and the angular components reverse sign at $(r, -\theta, z, t)$. Changing the angular integration variable therefore proves

$$J_2^\sharp = -J_2, \quad M_2^\sharp = -M_2, \quad J_1^\sharp = J_1, \quad M_1^\sharp = M_1.$$

An explicit transverse-curl example shows why compactness and nonzero angular frequency alone do not force either moment to vanish. This example tests precisely those structural properties, and is not a replacement for the retained source state. Let H be any nonzero real $C_c^\infty((r_-, r_+))$ function with $0 < r_- < r_+$, let $g \in C_c^\infty(\mathbb{R})$ be nonconstant, let n be a nonzero integer, and put $f = rH'$. Use zero old state and the real potential

$$\mathcal{A}_r = fg \sin(n\theta), \quad \mathcal{A}_\theta = 0, \quad \mathcal{A}_z = Hg \cos(n\theta).$$

It has the required transverse representation: $\phi = n\theta$, $\xi = (n/r)e_\theta$, $C = (-ifg, 0, Hg)$ in cylindrical components, $a = i\xi \times C$, and $C = i\xi \times a/|\xi|^2$ because $\xi \cdot C = 0$. The curl is exactly

$$v_r = -\frac{nHg}{r} \sin(n\theta), \quad v_\theta = fg' \sin(n\theta) - H'g \cos(n\theta), \quad v_z = -\frac{nfg}{r} \cos(n\theta).$$

Consequently

$$M_2 = n g g' \int_0^\infty r^2 (H')^2 dr, \quad M_1 = n^2 g g' \int_0^\infty r (H')^2 dr.$$

For example $g(z) = \exp(-1/(1 - z^2))$ on $|z| < 1$, extended by zero, has $g(z)g'(z) \neq 0$ at every $0 < |z| < 1$. Both integrals are strictly positive: a compactly supported smooth function with $H' = 0$ everywhere would be constant and hence zero. Thus both moments can be nonzero, with their displayed signed values, within the exact transverse-curl architecture.

18.4 Bounds in the actual fields and original chart units

At fixed (z, t) set

$$\|a\|_e^2 = \int_0^\infty r^e \langle |a|^2 \rangle dr, \quad e = 1, 2.$$

When an old-field norm is used below it can be restricted to the enclosing radial support of v and $\partial_z v$. This avoids imposing any decay condition on an unused exterior part of the old field. The following component bounds contain only actual fields:

$$|J_2| \leq \|w_{*,z}\|_2 \|v_\theta\|_2 + \|w_{*,\theta}\|_2 \|v_z\|_2 + \|v_z\|_2 \|v_\theta\|_2, \quad (59)$$

$$|J_1| \leq 2\|w_{*,z}\|_1 \|v_z\|_1 + \|v_z\|_1^2, \quad (60)$$

$$\begin{aligned} |M_2| &\leq \|\partial_z w_{*,z}\|_2 \|v_\theta\|_2 + \|w_{*,z}\|_2 \|\partial_z v_\theta\|_2 + \|\partial_z v_z\|_2 \|w_{*,\theta}\|_2 \\ &\quad + \|v_z\|_2 \|\partial_z w_{*,\theta}\|_2 + \|\partial_z v_z\|_2 \|v_\theta\|_2 + \|v_z\|_2 \|\partial_z v_\theta\|_2, \end{aligned} \quad (61)$$

$$|M_1| \leq 2(\|\partial_z w_{*,z}\|_1 \|v_z\|_1 + \|w_{*,z}\|_1 \|\partial_z v_z\|_1 + \|v_z\|_1 \|\partial_z v_z\|_1). \quad (62)$$

Each is the Cauchy–Schwarz inequality on the product measure $r^e dr d\theta/(2\pi) dY$, after applying the product rule in (56)–(57). For example, the derivative of $2w_{*,z}v_z + v_z^2$ is $2(\partial_z w_{*,z})v_z + 2w_{*,z}\partial_z v_z + 2v_z\partial_z v_z$, proving every coefficient in (62). For a (z, t) multi-index I , every further derivative is exactly obtained by replacing each product ab in J_e by

$$\partial^{I+(1,0)}(ab) = \sum_{K \leq I+(1,0)} \binom{I+(1,0)}{K} (\partial^K a)(\partial^{I+(1,0)-K} b)$$

inside its original radial integral. This identity follows inductively from the one-derivative product rule, and gives the corresponding finite-order moment bound by applying the same Cauchy–Schwarz inequality separately to every displayed summand.

The field v in these bounds is not an abstract replacement amplitude. For each actual band and harmonic write

$$\begin{aligned} t_j &= \chi_j B_j L_j y_j, & c_j &= \frac{i n_{\Phi,j} \times t_j}{k_{\beta_j} m_j |n_{\Phi,j}|^2}, & b_j &= t_j + \text{curl}_{*,\text{coeff}} c_j, \\ v &= \sum_j \text{Re}(Q_{\beta_j}^{-A} b_j e^{ik_{\beta_j} m_j \Phi_j}), & A &= \frac{1}{2} + h, & D &= \frac{1}{2} - h. \end{aligned} \quad (63)$$

Here $\text{curl}_{*,\text{coeff}}$ differentiates the chart coefficient, including the cutoff, auxiliary chain rule and cylindrical basis, but not the displayed exponential. This is exactly the remainder of the physical potential $Q_{\beta_j}^{1/2-A} c_j e^{ik_{\beta_j} m_j \Phi_j}$: physical curl introduces the factor $Q_{\beta_j}^{-1/2}$, and differentiating the exponential produces t_j by the transverse triple-product identity. Explicitly, on the absolute auxiliary torus in a fixed band,

$$\mathcal{D}_R = \partial_R + \sqrt{Q} dr r^{d_r-1} \mathbf{v}_r \cdot \partial_Y, \quad r = \sqrt{Q} R, \quad \mathcal{D}_Z^* = \varepsilon \partial_Z.$$

For the actual coefficient $c = (c_r, c_\theta, c_z)$, which has $\partial_\theta c = 0$, the operator is exactly

$$\text{curl}_{*,\text{coeff}} c = (-\varepsilon \partial_Z c_\theta, \varepsilon \partial_Z c_r - \mathcal{D}_R c_z, (\mathcal{D}_R + R^{-1}) c_\theta).$$

Coefficients originally on a band auxiliary torus are pulled back to the absolute torus before applying this formula; this retains the integer-covering chain derivative in $\partial_Y c$. In the same fixed chart the exact axial derivative is

$$\partial_z v = \sum_j \text{Re} \left[Q_{\beta_j}^{-A-D} (\partial_{Z_j} b_j + ik_{\beta_j} m_j (\partial_{Z_j} \Phi_j) b_j) e^{ik_{\beta_j} m_j \Phi_j} \right]. \quad (64)$$

The band power is fixed during chart differentiation and $\partial_z = Q_{\beta_j}^{-D} \partial_{Z_j}$; auxiliary phase evaluation adds no axial derivative because of (49). Thus (63)–(64) substitute the actual retained pair and all its curl terms into the bounds. For example, writing $\|b\|_{e,\beta}^2 = \int R^e \langle |b|^2 \rangle dR$ in that band, the triangle inequality and $|\text{Re } z| \leq |z|$ give

$$\|v\|_e \leq \sum_j Q_{\beta_j}^{(e+1)/4-A} \|b_j\|_{e,\beta_j}, \quad (65)$$

$$\|\partial_z v\|_e \leq \sum_j Q_{\beta_j}^{(e+1)/4-A-D} \|\partial_{Z_j} b_j + ik_{\beta_j} m_j (\partial_{Z_j} \Phi_j) b_j\|_{e,\beta_j}. \quad (66)$$

The exponent $(e+1)/4$ is exactly the square root of the radial measure factor $r^e dr = Q^{(e+1)/2} R^e dR$. Distinct bands and harmonics remain distinct in these sums. No limiting estimate as $q \downarrow 0$ has been inferred from the finite compact-patch norms.

For a single fixed chart, put $F_e = Q^{-2A-1/2}E_e^*$ as in the covariance bridge and $S^* = Q^{2A}S$. Direct change of radial variable gives

$$M_e = Q^{e/2-2A}M_e^*, \quad M_e^* = \int R^e E_e^* dR, \quad M_e^* = \varepsilon \partial_Z \int R^e S_{z,\tau_e}^* dR, \quad \varepsilon = Q^h,$$

where $\tau_2 = \theta$ and $\tau_1 = z$. Indeed the axial flux has factor $Q^{(e+1)/2-2A}$, its axial derivative contributes Q^{-D} , and dividing by $Q^{e/2-2A}$ leaves $Q^{1/2-D} = Q^h$. This proves the precise relation to the source chart moment without altering its axial anisotropy.

18.5 Exact relation to moments after phase evaluation

Let

$$\mathcal{S} = \langle w_* \otimes v + v \otimes w_* + v \otimes v \rangle_\theta, \quad \mathcal{S}^\circ = \mathcal{S} - \langle \mathcal{S} \rangle_Y.$$

Thus $\langle \mathcal{S} \rangle_Y = S$ and $\langle \mathcal{S}^\circ \rangle_Y = 0$. For the evaluated physical fields, denote the angular-only residual moments by M_e^{ev} . The proof of (50) holds after evaluation. The physical divergence, vector Laplacian and gradient identities therefore prove, without any auxiliary averaging,

$$M_e^{\text{ev}} = \partial_z \int_0^\infty r^e \mathcal{S}_{z,\tau_e}(r, z, t, Y(r, t)) dr.$$

The physical radial boundary terms still vanish by compact support; the chain rule is included in the physical radial derivative before that integration. Subtracting (56)–(57) proves the exact comparison

$$M_e^{\text{ev}} - M_e = \partial_z \int_0^\infty r^e \mathcal{S}_{z,\tau_e}^\circ(r, z, t, Y(r, t)) dr. \quad (67)$$

For example its absolute value is bounded by $\int r^e \sup_Y |\partial_z \mathcal{S}_{z,\tau_e}^\circ| dr$, because $Y(r, t)$ is independent of z . Zero auxiliary Haar average does not make the right side of (67) zero at the physical phase. The displayed map supplies precisely the remaining auxiliary-oscillation flux; it is not silently identified with the source fast mean.